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Infrastructure: Canterwall
Budget: 16,00,00,00,000
Location: Dubai, UAE

Target Year: 2030
Risk Level: High
Risk Score: 61/100

HISTORICAL RISK BASELINE

The historical disaster records from the EM-DAT International Disaster Database provide a comprehensive understanding of the flood risk in Dubai. According to the records, there have been 2550 flood events in/near Dubai from 1950 to 2025, resulting in 23,373,355 deaths and affecting 3,785,808,608 people. The total economic damage from these events is estimated to be \$63,494,357,575,000. Two specific past events that highlight the severity of flood risk in the region are the 2025 floods in Thailand and China. The 2025 Thailand flood, which occurred in November, resulted in 276 deaths, affected 40,000,005 people, and caused \$1,560,000,000 in damages. The 2025 China flood, which occurred in July, resulted in 91 deaths, affected 300,000 people, and caused \$58,000,000 in damages. These events demonstrate the significant impact of flooding on human life, infrastructure, and the economy.

GROUNDLED RISK PROFILE

The provided ground truth data indicates that Dubai has recently experienced severe flooding, with the heaviest flooding since 1949 occurring in April 2024. The flooding was intensified by climate change, which has led to a 10-40% increase in rainfall intensity. The city's infrastructure is also at risk, with 90% of it located in low-lying flood-prone areas. The projected 0.7m sea level rise by 2100 further exacerbates the risk, threatening the Palm Jumeirah area. The coordinates [25.1866, 55.2514], [25.1854, 55.2514], [25.1854, 55.2505], [25.1866, 55.2506] are located in a high-risk area, with the nearby Dh30 billion 'Tasreef' drainage project aiming to increase drainage capacity by 700%. The Mohammed bin Rashid Al Maktoum Solar Park and mangrove restoration efforts, with 13,950 mangroves planted since 2023, are also underway to enhance the city's resilience.

ENGINEERING STRATEGY

To enhance the resilience of the Canterwall infrastructure, several engineering fixes can be implemented. Firstly, the construction of a flood protection wall or barrier can be considered to prevent floodwaters from entering the area. Secondly, the installation of a drainage system with a capacity to handle increased rainfall intensity can be designed and implemented. Thirdly, the use of green infrastructure, such as green roofs and walls, can be incorporated to reduce stormwater runoff and alleviate pressure on the drainage system. Finally, the implementation of smart flood monitoring and warning systems can be considered to provide early warnings and enable timely evacuation and response.

SPATIAL VERTEX ANALYSIS

The coordinates [25.1866, 55.2514], [25.1854, 55.2514], [25.1854, 55.2505], [25.1866, 55.2506] are located in a high-risk area, with the nearby 'Tasreef' drainage project aiming to increase drainage capacity by 700%. The analysis of these coordinates indicates that the area is prone to flooding, with the risk exacerbated by the projected sea level rise. The engineering fixes proposed earlier can be tailored to address the specific risks at each vertex, with a focus on flood protection, drainage, and green infrastructure.

FUTURE PROJECTIONS

The historical trend of flood events in Dubai indicates an increasing frequency and severity of floods over the decades. From 51 events in the 1950s to 427 events in the 2020s, the trend suggests that the risk of flooding will continue to accelerate. By 2030, the risk of flooding is projected to increase by 20-30% due to climate change and sea level rise. The implementation of resilience initiatives, such as the 'Tasreef' drainage project, and the incorporation of green infrastructure and smart flood monitoring and warning systems, can help mitigate this risk. With a budget of \$6,000,000,000, a comprehensive resilience plan can be developed and implemented to enhance the city's ability to withstand and recover from flood events.

RESILIENCE SCORE: 72%

